



From water deficit to growth suppression: meta-analytic evidence for a functional hierarchy of drought adaptation in *Hippophae rhamnoides*

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Abstract

Drought constrains shrub establishment and persistence in arid and semi-arid restoration systems. Although *Hippophae rhamnoides* is widely planted in northern China, its drought responses vary among studies, hindering the identification of stable indicators. We synthesized 3,855 treatment–control comparisons from 49 controlled experiments across 12 Chinese provinces using random-effects meta-analysis and structural equation modeling. Results showed that drought adaptation in *H. rhamnoides* follows a functional cascade. Water status and membrane stability were the most consistent early constraints, followed by strong suppression of photosynthetic and gas-exchange traits. Photosynthesis acted as a key mediating factor linking water deficit, hormonal adjustment, and growth reduction. Under severe drought, membrane permeability increased by 89.5%, while leaf water content, net photosynthetic rate, transpiration rate, and stomatal conductance decreased by 37.2, 83.3, 91.2, and 92.9%, respectively. Therefore, drought evaluation should prioritize water status, membrane stability, photosynthetic performance, and growth traits.

Keywords: abscisic acid; gas exchange; membrane permeability; osmotic adjustment; physiological traits; shrub restoration; water deficit.

Introduction

Extreme drought events have become more frequent and intense under ongoing climatic warming (Tang *et al.* 2025). In arid and semi-arid regions, drought limits vegetation growth, community stability, and agricultural production, and can further alter ecosystem structure and function (Gao *et al.* 2021). As water availability becomes more variable, restoration and management require a clearer understanding of species-specific physiological responses and drought thresholds along water-availability gradients. Such information provides a quantitative basis

for evaluating vegetation performance in drought-prone regions (Du *et al.* 2025).

Hippophae rhamnoides is a deciduous shrub widely distributed in arid and semi-arid regions of northern China, and it is characterized by cold tolerance, drought tolerance, and a well-developed root system (Gan *et al.* 2025). Because of its nitrogen-fixing ability and its capacity to improve soil physicochemical properties, *H. rhamnoides* has been used in desertification control, slope protection, and soil-water conservation, particularly in regions such as Inner Mongolia and Xinjiang (He *et al.* 2015). A clearer synthesis of its drought-response traits is therefore needed

Highlights

- Water status and membrane stability are early drought constraints
- Severe drought strongly limits gas exchange and growth
- Osmolytes increase while growth-related hormones decline

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Abbreviations: ABA – abscisic acid; CAT – catalase; CFI – comparative fit index; C_i – intercellular CO₂ concentration; E – transpiration rate; F_v/F_m – maximal quantum yield of PSII photochemistry; GA – gibberellin; g_s – stomatal conductance; IAA – indole-3-acetic acid; lnRR – natural-log response ratio; MDA – malondialdehyde content; NPQ – nonphotochemical quenching; P_N – net photosynthetic rate; POD – peroxidase; Pro – proline; q_p – photochemical quenching coefficient; SEM – structural equation modeling; SOD – superoxide dismutase; SRMR – standardized root mean square residual; WUE – water-use efficiency; ZT – zeatin.

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to support more rational deployment and management of *H. rhamnoides* in arid regions (Lyu *et al.* 2022).

Under drought stress, ROS accumulation in *H. rhamnoides* leaves can induce oxidative pressure, promote membrane lipid peroxidation, and impair the photosynthetic apparatus (Chen *et al.* 2024a). Antioxidant enzymes, including superoxide dismutase (SOD), peroxidase (POD), and catalase (CAT), are involved in ROS scavenging and may reduce drought-induced oxidative injury (Zhang *et al.* 2021a). Malondialdehyde (MDA) content is commonly used to indicate the degree of membrane lipid peroxidation under stress (Du *et al.* 2025). In parallel, *H. rhamnoides* can accumulate osmolytes, such as proline and soluble sugars, which contribute to cellular osmotic balance and structural stability. These responses may help maintain photosynthetic function within a certain range (Zhang *et al.* 2021a).

At the morphological level, *H. rhamnoides* may reduce leaf area, adjust leaf number and spatial arrangement, and increase root length or the root-to-shoot ratio. These changes can reduce transpirational water loss and improve access to deeper water resources (Guo *et al.* 2007). In long-term drought experiments, 120 d of continuous stress reduced seedling height, stem diameter, and total biomass, whereas taproot length and the root-to-shoot ratio increased. This pattern suggests a shift in biomass allocation toward the root system under water deficit (Liu *et al.* 2024). Antioxidant enzyme activities also vary with drought intensity. Under moderate drought, SOD, POD, and CAT activities often increase, whereas under severe drought, decreases in SOD and POD activities have also been reported, sometimes together with relatively higher CAT activity (Chen *et al.* 2024a, Fang *et al.* 2024). These results indicate that drought responses in *H. rhamnoides* include both morphological adjustment and physiological regulation, but the direction and magnitude of specific traits may differ among studies.

Although drought responses of *H. rhamnoides* have been examined from the perspectives of antioxidant systems, photosynthesis, morphological traits, and hormonal regulation (Guo *et al.* 2007, Zhang *et al.* 2021a; Chen *et al.* 2024a, Yang *et al.* 2024), the available evidence remains difficult to compare directly. Studies differ in the definitions of drought intensity and duration, as well as in cultivation conditions, genotypes, sampling time, and measured variables. As a result, both the direction and magnitude of drought effects on the same trait may differ among studies. For example, some studies reported that drought increased antioxidant enzyme activities and helped maintain photosynthetic capacity within a certain range (Chen *et al.* 2024a). In contrast, other studies found that under prolonged or severe drought, SOD activity declined and both photosynthetic rate and PSII efficiency were strongly suppressed (Liu *et al.* 2024, Yang *et al.* 2024). These inconsistencies limit a clear understanding of drought-response mechanisms in *H. rhamnoides* and make it difficult to identify practical indicators for water management and restoration. A systematic species-level synthesis of physiological and morphological responses along water-availability gradients is therefore still needed.

Meta-analysis integrates effect sizes from independent studies within a common statistical framework and is therefore suitable for evaluating inconsistent experimental results. It also allows overall responses to be estimated while accounting for between-study heterogeneity (Kanukula *et al.* 2024). This approach has been increasingly used in studies of tree stress tolerance and functional traits, but species-level syntheses of *H. rhamnoides* drought responses across regions and water regimes in China remain limited.

Against this background, we considered drought adaptation in *H. rhamnoides* as a set of linked functional responses rather than as independent changes in isolated traits. We expected that water deficit would first be reflected in plant water status and membrane stability, because these traits directly represent tissue dehydration and cellular injury. Changes in photosynthesis were then expected to integrate the effects of hydraulic limitation, stomatal regulation, pigment status, and photochemical performance. Hormonal adjustment and growth reduction were considered downstream responses that reflect the balance between stress tolerance and resource allocation. Antioxidant enzymes and osmotic compounds were treated as buffering responses whose direction and magnitude may depend on drought stage.

Using this framework, we compiled published drought-stress experiments on *H. rhamnoides* from representative arid and semi-arid regions of China and addressed three questions. First, which traits show the most consistent responses across drought intensities? Second, do antioxidant enzymes and osmotic compounds act as general drought-resistance indicators, or are they stage-dependent responses? Third, does photosynthesis occupy an integrative position linking water deficit, hormonal regulation, and growth suppression? By answering these questions, this study aims to identify robust drought-response indicators and to clarify the functional hierarchy underlying drought adaptation in *H. rhamnoides*.

Materials and methods

Study region and overview of experimental sources:

All experiments included in this study were conducted in China and covered 12 provinces, mainly in the arid, semi-arid, and seasonally dry regions of northern, northwestern, and northeastern China. These areas represent the core natural and cultivated range of *Hippophae rhamnoides* and coincide with regions that experience severe soil erosion and ecological degradation. In all experiments, potted *Hippophae rhamnoides* seedlings were used as the study material, and most trials were conducted under controlled conditions, such as in greenhouses or plastic tunnels. Drought intensity was manipulated by regulating soil water content or soil water potential, thereby minimizing the confounding effects of natural precipitation. Under these standardized conditions, the experiments captured the characteristic physiological responses of individual plants to water deficit that are representative of drought-prone regions.

Literature search, inclusion criteria, and data extraction: We searched publications published between 2001 and 2025 in *CNKI*, *Wanfang Data*, *Web of Science*, *Google Scholar*, and *X-MOL* using combinations of the keywords “drought”, “water stress”, “*Hippophae rhamnoides*”, “photosynthesis”, “antioxidant”, “morphology”, and “hormone”. Studies were screened using the following criteria: (1) experiments were conducted in China and used *Hippophae rhamnoides* as the experimental material; (2) experiments were pot trials conducted under controlled conditions (e.g., greenhouses or plastic tunnels) to avoid confounding effects of natural precipitation on soil-water treatments; (3) well-watered plants served as controls, and drought treatments were imposed by reducing soil water potential or soil water content; soil water status was reported quantitatively as water potential (kPa) or volumetric/gravimetric soil water content (%); (4) studies reported, for both control and drought treatments, sample size, mean, and either standard deviation (SD) or standard error (SE); (5) each treatment included at least three replicates.

Drought duration was also evaluated during screening. However, it was not used as a mandatory inclusion criterion because it was reported inconsistently across the original studies, for example as treatment days, sampling stages, or endpoint descriptions, which made strict standardization difficult at the species level. To improve comparability across studies and avoid introducing additional classification bias, we prioritized

soil water potential and soil water content as the primary indicators of drought treatment intensity. Variation in drought duration was therefore treated as a potential source of heterogeneity rather than a fixed screening threshold.

When duplicate publications or clearly overlapping datasets were identified, only the study providing the most complete information was retained. In total, 65 publications were initially retrieved. Studies that did not meet the criteria or lacked usable data were excluded, resulting in 49 publications (Li 2001, Li *et al.* 2001, 2002, 2006, 2017; Han *et al.* 2002, Liu 2005, Liu *et al.* 2005, 2006, 2017, 2022, 2025a; Wang 2006, 2018; Hou 2007, Xu 2007, Gao *et al.* 2009, Xu *et al.* 2009, 2015; Cui 2010, 2017; Cui *et al.* 2010, Gao 2010, 2018; Zhang 2010, 2015; Cai *et al.* 2011, Fang *et al.* 2012, Ma *et al.* 2013a,b,c; 2014; Pei *et al.* 2013, Wu *et al.* 2013, Song 2016, Sun *et al.* 2018, Chen *et al.* 2019, 2024b; Nan *et al.* 2019, 2021; Mou 2020, Zhang *et al.* 2020, 2021c; Hu *et al.* 2021, Tang *et al.* 2021, Ai 2024, Cheng *et al.* 2024, Tian 2024, Xia 2024) from 12 provinces (Fig. 1). From these studies, 3,855 treatment–control comparisons were extracted. When trait values were reported only graphically, data were digitized using *WebPlotDigitizer* (ver. 4.2). When only SE was reported, SD was calculated from SE and sample size.

$$SD = SE \times \sqrt{n}$$

Accordingly, SD was obtained. For the few cases in which neither SD nor SE was reported, SD was imputed as one-tenth of the mean value following previous studies;

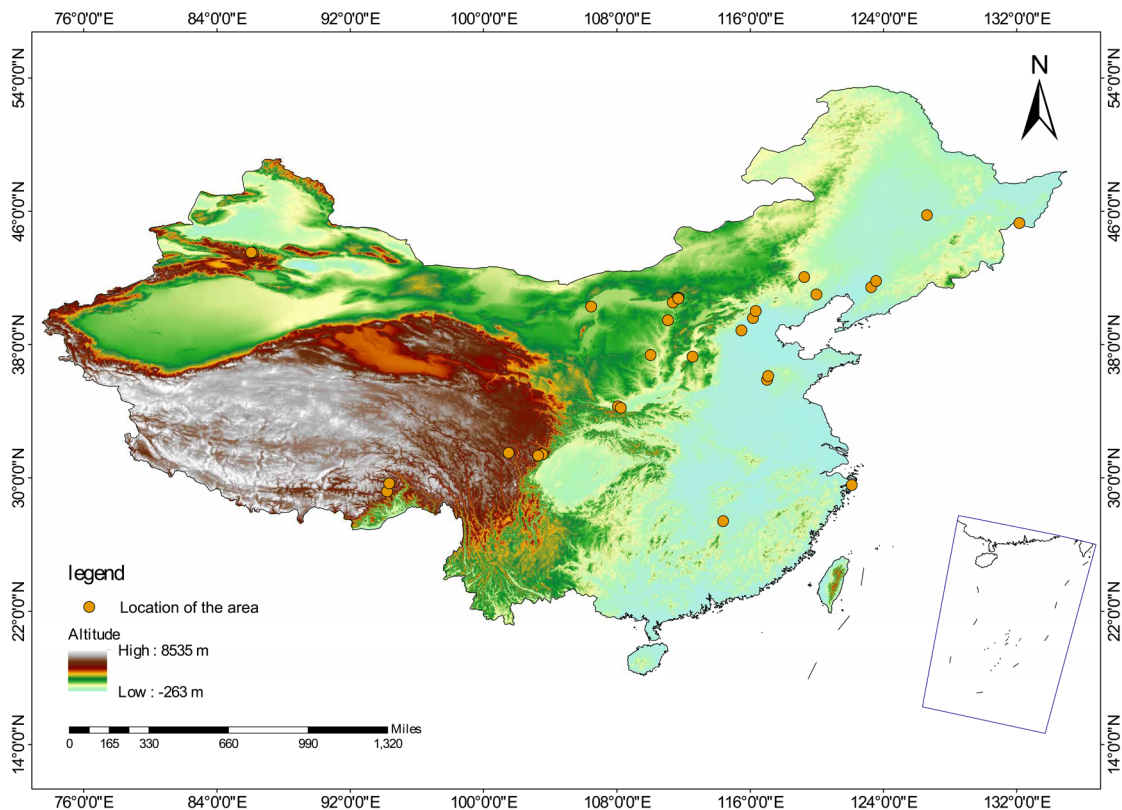


Fig. 1. The distribution of study points. The base map is made from the standard map of the Ministry of Natural Resources. The review number is GS (2023) 2767, and the map boundary is not modified.

these data points were subsequently treated as sensitivity cases in the analysis. The compiled study-level database used for the meta-analysis was provided in the *Supplement* to facilitate transparency and reproducibility (Luo *et al.* 2006).

A PRISMA(S1) flow diagram, the full list of included studies, and the compiled study-level database were provided in the *Supplement* to improve transparency and reproducibility.

Indicator functional module classification and drought-intensity definitions: Based on their functional roles in drought responses of *Hippophae rhamnoides*, the extracted variables were grouped into four categories. The four functional modules were defined as major response dimensions of drought adaptation for cross-study synthesis rather than as strictly independent physiological compartments. Because some variables may span conceptual boundaries, traits were assigned according to their dominant functional relevance under drought stress. For example, leaf water potential was classified under water status because it primarily reflects plant water relations, whereas membrane permeability and relative electrical conductivity were grouped under membrane stability. After all, they mainly indicate stress-induced membrane injury and integrity loss. Antioxidant defense and osmotic adjustment were combined into one composite module because, although mechanistically distinguishable, they are closely coordinated under drought stress and jointly contribute to cellular homeostasis, osmotic balance, and stress buffering (Kolyva *et al.* 2025).

(1) Antioxidant and osmotic adjustment: activities of SOD, POD, and CAT; MDA content; membrane permeability; relative electrical conductivity; leaf water potential; leaf water content; proline, soluble sugar, and soluble protein concentrations;

(2) Phenotype and growth: plant height, leaf area, and biomass (root and total biomass);

(3) Photosynthesis: maximum photochemical efficiency of PSII (F_v/F_m), photosynthetic rate, transpiration rate, stomatal conductance, stomatal limitation value, intercellular CO₂ concentration, photochemical quenching, nonphotochemical quenching, electron transport efficiency, chlorophyll, and water-use efficiency;

(4) Hormone adjustment: zeatin (ZT), indole-3-acetic acid (IAA), gibberellins (*e.g.*, GA₃, GA₄₀), abscisic acid (ABA), and related phytohormones.

Compared with drought duration, soil water status was reported more consistently across studies and therefore served as the primary basis for drought-intensity classification in this meta-analysis. These thresholds were intended to define an approximately comparable drought gradient across studies rather than to imply complete equivalence of absolute soil water content among different experiments. Drought intensity was standardized into three levels according to soil water potential or soil water content reported in the original studies: mild water stress: soil water content approximately 51–70%; moderate water stress: soil water content approximately 35–50%; severe water stress: soil water content approximately 10–34%.

For studies that reported only relative soil water content or irrigation ratios, soil water status was converted to the above categories using local field capacity information. When a reasonable conversion was not possible, the corresponding data were included only in the overall effect-size analysis and excluded from comparisons among drought-intensity classes.

Effect-size calculation, meta-analysis, and structural equation modeling: Effects of drought treatments on each variable were quantified using the natural-log response ratio (lnRR), following Hedges *et al.* (1999) and standard procedures in ecological meta-analysis. For each treatment–control comparison, lnRR was calculated as follows:

$$RR = \ln(X_t/X_c) = \ln(X_t) - \ln(X_c)$$

Here, X_t and X_c denote the mean values of the drought-treatment and control groups, respectively. Values of lnRR > 0 indicate that drought increased the metric, whereas lnRR < 0 indicates a decrease. The sampling variance (v) was calculated as follows:

$$v = \frac{SD_t^2}{N_t X_t^2} + \frac{SD_c^2}{N_c X_c^2}$$

In these equations, N_t and N_c denote the sample sizes of the treatment and control groups, respectively, and SD_t and SD_c denote the corresponding standard deviations. Each lnRR was weighted by the inverse of its variance (v), *i.e.*, $w = 1/v$, as follows:

$$w = \frac{1}{v} = \frac{1}{\frac{SD_t^2}{N_t X_t^2} + \frac{SD_c^2}{N_c X_c^2}}$$

Because differences in experimental conditions, drought treatment measures, measured traits, and research backgrounds were expected to generate substantial between-study heterogeneity, and the heterogeneity test showed that $I^2 > 50\%$, a random-effects model was used to estimate the pooled effect size.

The overall mean effect size (lnRR) was calculated as the weighted mean of individual lnRR values, with more precise estimates receiving greater weight. The weighted mean was computed as follows:

$$RR_{++} = \frac{\sum_{i=1}^m \sum_{j=1}^k w_{ij} RR_{++}}{\sum_{i=1}^m \sum_{j=1}^k w_{ij}}$$

Here, w_{ij} is the weighting factor for the j -th observation in the i -th group; m is the number of groups, and k is the number of treatment–control comparisons within group i . The weighted standard error of the overall mean effect size was calculated as follows:

$$SD(RR_{++}) = \sqrt{\frac{1}{\sum_{i=1}^m \sum_{j=1}^k w_{ij}}}$$

Statistical significance was assessed using 95% confidence intervals (CIs). An effect was considered

significant when the 95% CI of lnRR did not include zero (Hedges *et al.* 1999). Negative effects were inferred when the upper CI bound was < 0 , whereas positive effects were inferred when the lower bound was > 0 . The CI was calculated as follows:

$$95\%CI = RR_{++} \pm 1.96SD(RR_{++})$$

All analyses were performed in *R* (ver. 4.4.3). Effect sizes (lnRR) and 95% confidence intervals (CIs) were estimated using the *metafor* package under a random-effects framework, and forest plots were generated with *ggplot2* and *forestplot*. Between-study heterogeneity was quantified using I^2 statistics. Cases lacking variance information and requiring SD imputation were retained as sensitivity cases and were considered cautiously during interpretation. A correlation matrix based on meta-analytic effect sizes among the 26 retained variables. Only variables with sufficient complete and pairwise comparable observations were included. Traits with excessive missing values or insufficient sample size were excluded to avoid unstable correlation estimates.

Correlation structures were synthesized from meta-analytic effect sizes among the retained variables, and traits with excessive missing values or insufficient paired observations were excluded from the correlation matrix and subsequent SEM fitting. For the SEM analysis, variables were first screened according to data completeness and comparability across studies. Because not all studies reported all traits, the SEM was not built from a complete raw-data matrix covering all 49 studies and 30 indicators.

Instead, it was constructed from selected variables with sufficient information to support integrated analysis.

SEM was conducted using *lavaan* and *piecewiseSEM* (CFI=0.926, SRMR=0.072) to characterize the coordinated relationships among drought intensity and the four functional modules. Given the sparsity of the original study-trait matrix, the SEM should be interpreted as an exploratory and hypothesis-generating analysis rather than a strict causal model. Model fit was evaluated using standard SEM fit indices, and key indices are reported in the Results section. To improve transparency and reproducibility, the compiled meta-analysis database and supplementary data tables were provided as supplementary materials.

Results

Species-level effects (lnRR) of functional modules across drought intensities: The trait-level lnRR responses across mild, moderate, and severe drought are summarized in Fig. 2. Missing trait-intensity combinations indicate that no comparable data were available for that drought class in the retained dataset.

Responses and variability of membrane stability and water-status traits: Across mild (LD), moderate (MD), and severe (SD) drought intensities, indices of water status and membrane stability in *Hippophae rhamnoides* exhibited a progressive, intensity-dependent response. Overall, increasing drought intensity amplified changes

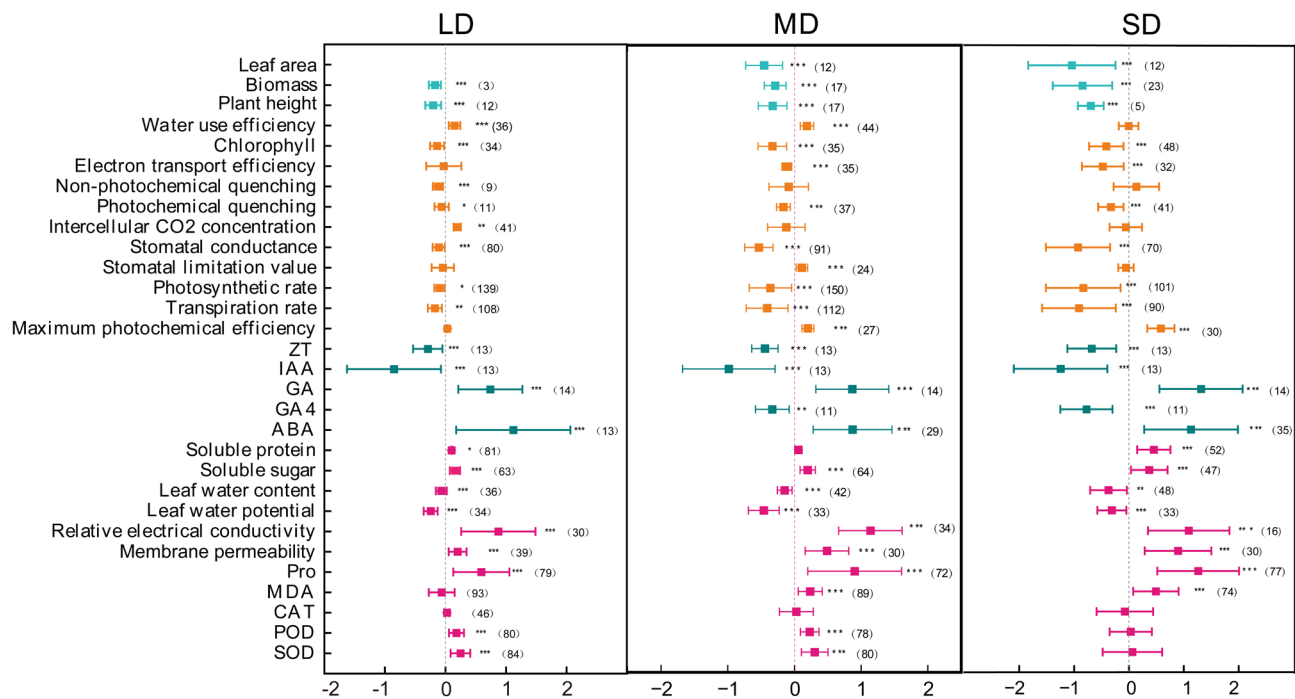


Fig. 2. Natural-log response ratios (lnRRs; weighted means $\pm$ 95% confidence intervals [CIs]) of the measured traits under mild drought (LD), moderate drought (MD), and severe drought (SD). Positive lnRR values indicate increases relative to the well-watered control, whereas negative values indicate decreases. Traits are grouped into four functional modules: Antioxidant and osmotic adjustment, Photosynthesis, Hormonal adjustment, and Phenotype and growth. Missing trait-intensity combinations indicate that no comparable data were available for the corresponding drought class. Colors indicate the four functional modules.

in membrane-stability metrics and drove sustained deterioration in water-status traits. However, several variables exhibited nonlinear responses or partial stabilization under severe drought. For membrane stability, membrane permeability increased by 19.9% under LD, indicating the onset of membrane impairment. The increase reached 48.7% under MD and rose further to 89.5% under SD, representing the strongest response across intensities and indicating pronounced membrane damage. Consistent with this pattern, relative electrical conductivity increased by 87.4% under LD and by 113.8% under MD, indicating substantial electrolyte leakage. Under SD, relative electrical conductivity decreased slightly to 108.8%. Although it did not continue to increase, values remained high, suggesting that this metric approached an upper bound under severe drought and that the response began to plateau. Water-status traits showed a similar pattern of deterioration. Leaf water content declined modestly by 6.9% under LD and by 14.6% under MD, indicating an increasingly severe water deficit. Under SD, leaf water content decreased by 37.2%, reflecting pronounced dehydration and reduced capacity to maintain tissue hydration. Consistent with these changes, leaf water potential became more negative under LD (-0.2446) and further decreased under MD (-0.4605), indicating progressively increasing water stress. Under SD, leaf water potential was -0.3118 . The value remained low, but the change was smaller than under MD, suggesting partial stabilization of plant water status at the highest drought intensity. Taken together, membrane permeability and electrolyte leakage generally increased with drought intensity, whereas leaf water status deteriorated overall. Notably, both relative electrical conductivity and leaf water potential showed nonlinear responses or attenuated changes under SD.

Responses of photosynthetic and gas-exchange traits across drought intensities: Across LD, MD, and SD drought intensities, photosynthetic quenching and gas-exchange traits showed an overall decline in carbon assimilation and water fluxes, accompanied by coordinated shifts in photochemistry and energy-partitioning parameters. Responses were heterogeneous across traits. Some variables changed monotonically with increasing drought severity, whereas others were nonmonotonic or reversed direction from moderate to severe drought, enabling an integrated interpretation across drought stages. Photochemistry and energy partitioning: The $\ln RR$ of maximum PSII photochemical efficiency (F_v/F_m) was 0.0311 under LD, increased to 0.2002 under MD, and further increased to 0.5822 under SD, indicating a pronounced drought-associated shift in this trait. Photochemical quenching declined progressively with drought ($\ln RR = -0.063$, -0.1656 , and -0.3294 under LD, MD, and SD, respectively), suggesting that a smaller fraction of absorbed energy was allocated to photochemical reactions. Nonphotochemical quenching decreased under LD and MD ($\ln RR = -0.1245$ and -0.0864 , respectively) but increased under SD ($\ln RR = 0.135$), indicating a reversal from mild-moderate to severe drought. Electron transport efficiency decreased stepwise with drought

($\ln RR = -0.0289$, -0.1141 , and -0.4772), consistent with progressively stronger inhibition of electron transport. Chlorophyll content also declined across the gradient ($\ln RR = -0.1388$, -0.3318 , and -0.4122), implying reduced pigment pools and diminished capacity for light capture and utilization with increasing drought intensity. Gas exchange and carbon assimilation: Transpiration rate declined by 17.6% under LD, 41.1% under MD, and 91.2% under SD, indicating progressively stronger suppression of water loss with increasing drought severity. Photosynthetic rate decreased by 9.7, 36.13, and 83.3% under LD, MD, and SD, respectively, indicating a sustained reduction in carbon assimilation that approached minimal levels under severe drought. Stomatal conductance decreased by 17.3% (LD), 53.5% (MD), and 92.9% (SD), demonstrating a marked drought-induced restriction of stomatal opening and gas exchange. In contrast, the stomatal limitation value showed a nonmonotonic response ($\ln RR = -0.0441$, 0.1112, and -0.053 under LD, MD, and SD, respectively). Intercellular CO_2 concentration increased by 19.4% under LD but shifted in the opposite direction under MD and SD ($\ln RR = -0.1226$ and -0.0573), again indicating a nonmonotonic response. Water-use efficiency increased under LD and MD ($\ln RR = 0.149$ and 0.1872) but was essentially unchanged under SD ($\ln RR = -0.0047$), suggesting that the apparent enhancement under mild-moderate drought was not maintained under severe drought. Overall, increasing drought severity drove persistent declines in transpiration, photosynthetic rate, and stomatal conductance, with the largest reductions under SD. By contrast, intercellular CO_2 concentration, stomatal limitation value, and nonphotochemical quenching exhibited nonmonotonic or direction-changing responses, implying that stomatal constraints and energy-dissipation processes may operate differently across drought stages.

Responses and variability of antioxidant and osmotic-adjustment traits under drought: Across drought intensities, antioxidant and osmotic-adjustment traits exhibited greater variability and stronger stage dependence. In general, several antioxidant enzymes increased from mild to moderate drought but declined under severe drought, whereas osmolytes tended to accumulate with increasing drought intensity. By contrast, indices of oxidative damage and membrane destabilization increased with drought severity and reached high levels under SD, with some traits showing attenuation or partial plateauing. The $\ln RR$ of SOD activity was 0.2478, 0.3030, and 0.0632 under LD, MD, and SD, respectively, indicating an increase from mild to moderate drought followed by a pronounced decline under SD. A similar pattern was observed for POD activity ($\ln RR = 0.1814$, 0.2271, and 0.0333). CAT activity changed only slightly under LD and MD ($\ln RR = 0.0236$ and 0.0277) but became negative under SD ($\ln RR = -0.0742$), suggesting reduced CAT-mediated detoxification capacity at high drought intensity. Together, these results indicate that antioxidant enzyme activities did not follow a simple monotonic trend but instead varied across drought stages. Markers of oxidative damage and membrane instability increased

overall with drought severity. The lnRR of malondialdehyde (MDA) shifted from -0.0628 under LD to 0.2359 under MD and 0.4907 under SD, consistent with progressively stronger oxidative injury. Membrane permeability increased continuously across the gradient (lnRR = 0.1998 , 0.4870 , and 0.8949), indicating a pronounced increase in membrane leakiness. Relative electrical conductivity remained high (lnRR = 0.8737 , 1.1382 , and 1.0883) and decreased slightly under SD, consistent with a high-level plateau response. Water status deteriorated concurrently: leaf water content declined stepwise (lnRR = -0.0696 , -0.1461 , and -0.3723), indicating progressively intensified dehydration. Leaf water potential also shifted markedly across the gradient (lnRR = -0.2446 , -0.4605 , and -0.3118), further supporting drought-driven changes in plant water status. Collectively, deterioration in membrane integrity and water status paralleled increasing drought intensity and aligned with the overall increase in oxidative stress. Osmotic-adjustment compounds showed an overall accumulation trend. Pro increased from 0.5916 (LD) to 0.9016 (MD) and 1.2616 (SD) in lnRR, indicating sustained osmolyte accumulation under drought. Soluble sugars also increased (lnRR = 0.1563 , 0.1980 , and 0.3704). Soluble proteins exhibited a stage-dependent pattern (lnRR = 0.1007 , 0.0581 , and 0.4536), decreasing under MD and then increasing sharply under SD. Chlorophyll content declined continuously (lnRR = -0.1388 , -0.3318 , and -0.4122), consistent with earlier results and indicating sustained depletion of photosynthetic pigments with increasing drought severity.

Responses of hormonal regulation and growth phenotypes across drought intensities: Across drought intensities, hormone-related traits and growth phenotypes changed in parallel, and several variables exhibited nonmonotonic response patterns. Overall, some hormonal indicators shifted consistently with increasing drought, whereas phenotypic traits such as plant height generally showed stronger declines under higher drought intensity. Hormonal metrics responded heterogeneously to drought intensity. The lnRR of abscisic acid (ABA) was 1.1199 (LD), 0.8703 (MD), and 1.1313 (SD), showing a nonmonotonic pattern with an initial decrease followed by an increase under SD. GA₄ remained at -0.3332 under both LD and MD but declined further to -0.7744 under SD, indicating a stronger reduction in GA₄ under severe drought. GA increased across the gradient (lnRR = 0.7408 , 0.8662 , and 1.3148 under LD, MD, and SD, respectively). In contrast, indole-3-acetic acid (IAA) decreased progressively (lnRR = -0.8498 , -0.9841 , and -1.2461), and zeatin (ZT) also showed a sustained decline (lnRR = -0.3424 , -0.4437 , and -0.6767). These patterns indicate that hormonal regulation across drought intensities was not uniform in direction and displayed clear stage dependence. Growth phenotypes showed both monotonic and nonmonotonic responses. Plant height decreased under LD and MD (lnRR = -0.2033 and -0.3273), and it declined further under SD (lnRR = -0.6957), indicating that severe drought exerted a stronger inhibitory effect on height growth. Biomass decreased more strongly with increasing drought

intensity (lnRR = -0.1743 , -0.2898 , and -0.8476 under LD, MD, and SD), suggesting sustained suppression of overall growth and carbon accumulation. Leaf area decreased under MD and SD (lnRR = -0.4550 and -1.0409), with stronger reductions at higher drought intensity.

Shared response patterns across drought intensities:

Across the three drought intensities, *Hippophae rhamnoides* exhibited a highly coherent, unidirectional physiological trajectory. As drought intensified from mild to severe, tissue water-retention capacity declined continuously, accompanied by increased membrane permeability and progressive dehydration. This response represented the earliest and most stable foundational constraint on physiological functioning. Against this background, stomatal closure and suppressed transpiration drove coordinated downregulation of the photosynthesis–stomatal system. The dominant limitation shifted from primarily stomatal regulation under mild drought – suggesting largely reversible inhibition – to stronger nonstomatal constraints under moderate to severe drought, characterized by reduced pigment content and impaired electron transport together with accumulation of oxidative damage, consistent with structural impairment. Antioxidant capacity and osmotic adjustment were generally induced to buffer stress; however, under severe stress, enzymatic defenses appeared to approach saturation, and passive accumulation of damage markers and osmolytes became the dominant signature. In parallel, the hormonal network was reorganized in a directionally consistent manner, with stress signaling (centered on ABA) upregulated and growth signaling downregulated. These changes ultimately manifested as sustained declines in plant height, biomass, leaf area, and other indicators of growth and canopy expansion. Overall, the drought response in *Hippophae rhamnoides* reflects a system-level functional shift from “growth priority” to “defense priority”. Under severe drought, regulatory capacity appeared to approach an upper limit, such that defensive compensation could not offset the cascading suppression of photosynthesis and growth imposed by water deficit and membrane injury, resulting in a survival-maintenance strategy at the expense of strongly constrained growth.

Covariation among traits based on meta-analytic effect sizes:

The correlation matrix based on meta-analytic effect sizes indicates broadly consistent coordination and coupling among functional modules in *Hippophae rhamnoides* under drought stress (Fig. 3). Overall, the photosynthesis–stomatal–growth module showed the strongest internal associations, followed by coupling between growth-related hormones and morphological traits/biomass. In contrast, several defense and osmotic-adjustment variables tended to vary inversely with photosynthetic and growth traits. The photosynthesis–stomatal–growth module was strongly integrated. Photosynthetic rate was almost perfectly correlated with transpiration ($r \approx 0.96$). It was also positively associated with plant height ($r \approx 0.78$) and moderately positively correlated with water-use efficiency ($r \approx 0.35$). Transpiration rate was likewise

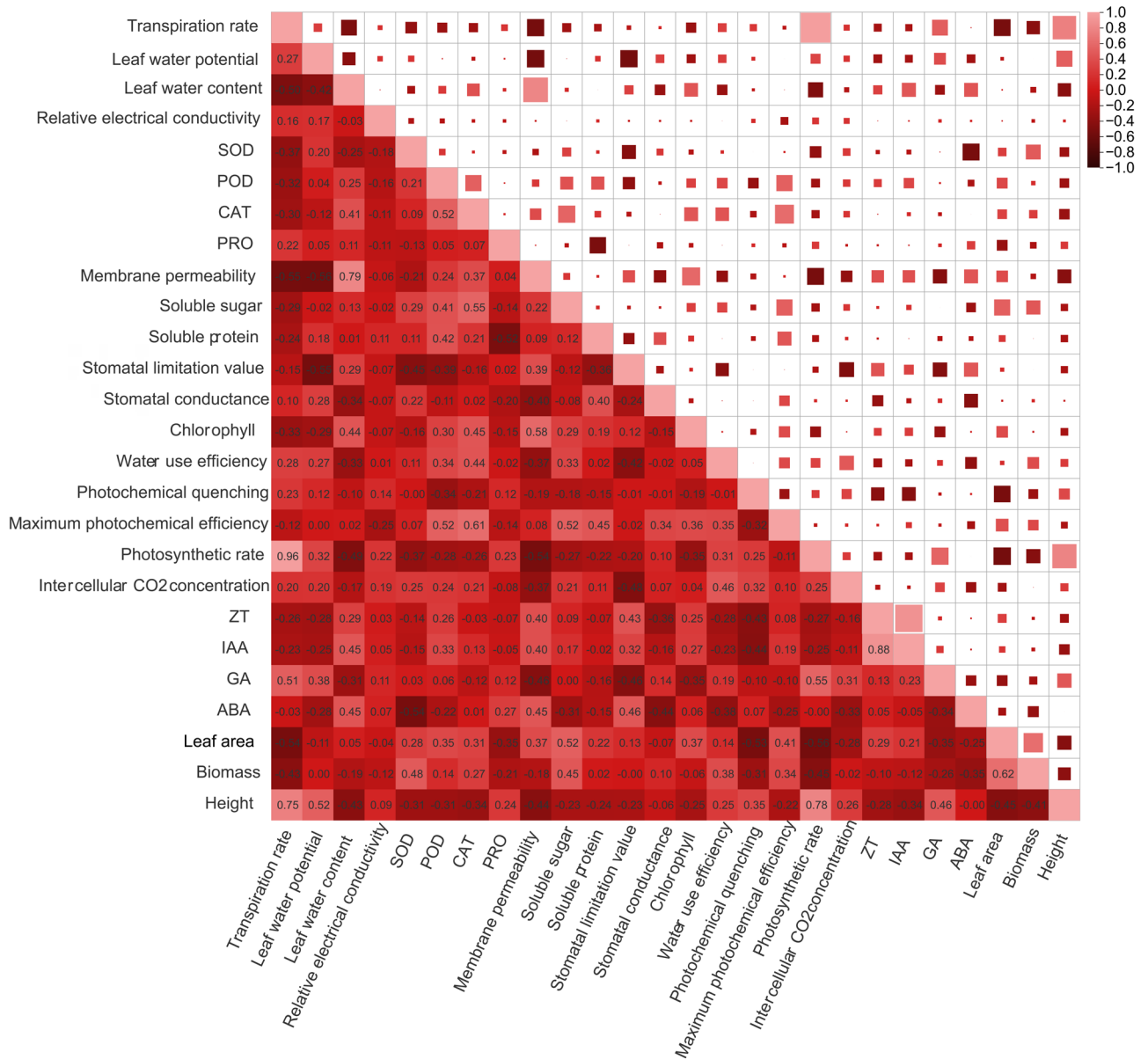


Fig. 3. Correlation matrix of meta-analytic effect sizes. The heatmap shows *Pearson's* correlation coefficients (r) among 26 standardized response variables. The color scale ranges from -1 to 1 ; the sign indicates the direction of the association, and the absolute value indicates its strength. SOD – superoxide dismutase; POD – peroxidase; CAT – catalase; Pro – proline; IAA – indole-3-acetic acid; ZT – zeatin; GA – gibberellin; ABA – abscisic acid.

positively correlated with plant height ($r \approx 0.75$) but was moderately negatively correlated with leaf relative water status ($r \approx -0.50$). Leaf area was positively correlated with total biomass ($r \approx 0.62$), indicating that canopy expansion broadly covaried with dry-matter accumulation. Hormonal signals exhibited directionally consistent coupling with photosynthesis and growth. GA content was positively correlated with photosynthetic rate ($r \approx 0.55$), transpiration ($r \approx 0.51$), and plant height ($r \approx 0.46$). ZT and IAA were strongly correlated ($r \approx 0.88$). Together with GA, these growth-related hormones tended to vary positively with photosynthetic rate, transpiration, plant height, leaf area, and biomass. This pattern indicates that, within

the compiled dataset, higher levels of growth-associated hormonal signals generally co-occurred with stronger photosynthetic performance and greater plant size. By contrast, ABA was negatively correlated with stomatal conductance ($r \approx -0.44$), leaf area ($r \approx -0.25$), and biomass ($r \approx -0.35$) and showed a moderate negative correlation with GA ($r \approx -0.34$). This pattern is consistent with the interpretation that elevated ABA under drought is generally associated with stomatal closure and reduced aboveground growth. Several defense and osmotic-adjustment variables showed clear inverse relationships with photosynthetic rate and growth traits. SOD activity was moderately negatively correlated with both photosynthetic

rate and transpiration rate ($r \approx -0.37$) and was negatively correlated with leaf area ($r \approx -0.54$). Pro content was also negatively correlated with leaf area and biomass ($r \approx -0.35$ and -0.21 , respectively). In addition, indicators reflecting membrane injury and dehydration (*e.g.*, relative electrical conductivity and leaf water content) tended to correlate negatively with photosynthetic and growth traits; for example, transpiration rate correlated at approximately $r \approx -0.50$ with relative electrical conductivity. Collectively, these patterns suggest that, within the drought intensities and evidence base represented here, increases in defense/osmotic-adjustment metrics and stress-damage indicators were typically accompanied by reduced photosynthetic capacity and constrained growth. These correlations should be interpreted cautiously. They were derived from meta-analytic effect sizes pooled across studies and are therefore influenced by between-study differences in sample size and missing variables. As such, they primarily reflect species-level covariation across the available evidence base, rather than causal relationships within a single experiment or at the level of individual plants.

Structural equation modeling of functional-module pathways

Direct and indirect effects of drought intensity on functional modules: The structural equation model of functional modules showed an overall acceptable fit to the data (CFI = 0.926, SRMR = 0.072), and the fitted coefficients were interpreted as statistical relationships among functional modules within an exploratory integrative framework (Fig. 4). Drought intensity exerted a strong positive standardized direct effect on the antioxidant–osmotic adjustment module ($\beta = 0.827$, $P < 0.01$) and a moderate positive direct effect on the phenotypic and growth adjustment module ($\beta = -0.416$, $P < 0.05$). In contrast, drought intensity had a significant negative direct effect on the photosynthesis module ($\beta = -0.762$, $P < 0.05$). The direct effect of drought intensity on the hormone-regulation module was negative but not significant ($\beta = -0.195$, $P > 0.05$). The path coefficient from the photosynthesis module to the hormone-regulation module was strongly negative ($\beta = -0.913$, $P < 0.01$). By comparison, paths linking the antioxidant–osmotic adjustment and phenotypic and growth adjustment modules to the photosynthesis and hormone-regulation modules were not significant (all $|\beta| < 0.5$, $P > 0.05$). Effect decomposition further indicated that the total effect of drought intensity on hormone regulation was dominated by an indirect pathway mediated by photosynthesis, whereas total effects on the antioxidant–osmotic adjustment and phenotypic and growth adjustment modules were driven primarily by direct paths. These indirect effects indicate that photosynthesis was not simply one drought-responsive trait among others. Within the fitted SEM, photosynthesis represented the main statistical bridge through which drought-related changes were linked to hormonal rebalancing. This result is consistent with the trait-level responses: drought strongly reduced stomatal conductance, transpiration, and net photosynthetic rate,

while hormone-related traits and growth traits also shifted with increasing drought intensity. Therefore, the central position of photosynthesis in the model reflects its close connection with both upstream water limitation and downstream physiological adjustment. However, because the SEM was constructed from pooled effect sizes across studies, these pathways should be interpreted as cross-study associations rather than direct evidence of causality.

This effect decomposition is important for interpreting the proposed hierarchy. The strong direct effect of drought intensity on the antioxidant and osmotic adjustment module indicates that dehydration and membrane injury responded closely to the imposed water deficit. The negative direct effect on photosynthesis shows that gas exchange and photochemical performance were strongly constrained as drought intensified. By contrast, the effect of drought intensity on hormone regulation was expressed mainly through the photosynthesis pathway, suggesting that hormonal changes were more closely linked to the physiological status of the plant than to the drought category alone. Therefore, the SEM does not prove a fixed causal sequence, but it supports a biologically plausible cascade in which water deficit and membrane injury constrain photosynthetic functioning, and changes in photosynthesis are associated with hormonal rebalancing and growth suppression.

Latent-construct specification and standardized factor loadings in the measurement model: To represent the integrated physiological response of *Hippophae rhamnoides* to drought within the structural equation model, 26 observed variables were grouped into four latent constructs: Antioxidant and osmotic adjustment, Photosynthesis, Hormone regulation, and Phenotype and growth (Table 1). Each construct was estimated using standardized factor loadings of its corresponding indicators. The sign and magnitude of each loading were interpreted as the direction and relative contribution of each indicator to the latent construct.

(1) Antioxidant and osmotic adjustment: This construct was defined by 10 indicators. Relatively high positive loadings were observed for membrane permeability (0.2875), leaf water content (0.2260), and soluble sugars (0.1330). Loadings for antioxidant enzyme activities were lower, including CAT (0.1358), POD (0.1091), soluble protein (0.0413), and SOD (0.0092). Relative electrical conductivity (-0.0582), leaf water potential (-0.2027), and Pro (-0.0730) loaded negatively. Overall, higher construct values primarily corresponded to increased membrane permeability, changes in tissue water status, and soluble sugar accumulation, together with lower leaf water potential. This pattern indicates that the construct mainly captured cellular dehydration, membrane destabilization, and osmotic buffering under drought.

(2) Photosynthesis: This construct integrated nine indicators spanning chlorophyll content and gas-exchange and photochemical traits. Photosynthetic rate showed the strongest loading (0.4446), followed by photochemical quenching (0.2315), intercellular CO₂ concentration (0.1925), stomatal conductance (0.1377), and water-

Table 1. Observed indicators and standardized factor loadings for the four latent constructs in the structural equation model (SEM). SOD – superoxide dismutase; POD – peroxidase; CAT – catalase; Pro – proline; IAA – indole-3-acetic acid; ZT – zeatin; ABA – abscisic acid; GA – gibberellin.

Antioxidant and osmotic adjustment		Photosynthesis		Hormone adjustment		Phenotype and growth	
Indicator	Standardized factor loading	Indicator	Standardized factor loading	Indicator	Standardized factor loading	Indicator	Standardized factor loading
SOD	0.0092	Chlorophyll	−0.2773	IAA	0.4430	Plant height	−0.5572
POD	0.1091	Photosynthetic rate	0.4446	ZT	0.4492	Leaf area	−0.4721
Pro	−0.0730	Stomatal conductance	0.1377	ABA	0.1552	Total biomass	−0.1852
Soluble sugar	0.1330	Intercellular CO ₂ concentration	0.1925	GA	−0.5997		
Membrane permeability	0.2875	Stomatal limitation value	−0.2569				
Relative electrical conductivity	−0.0582	Photochemical quenching	0.2315				
Leaf water potential	−0.2027	Maximum photochemical efficiency	−0.1793				
Leaf water content	0.2260	Water-use efficiency	0.1239				
CAT	0.1358	Transpiration rate	−0.3053				
Soluble protein	0.0413						

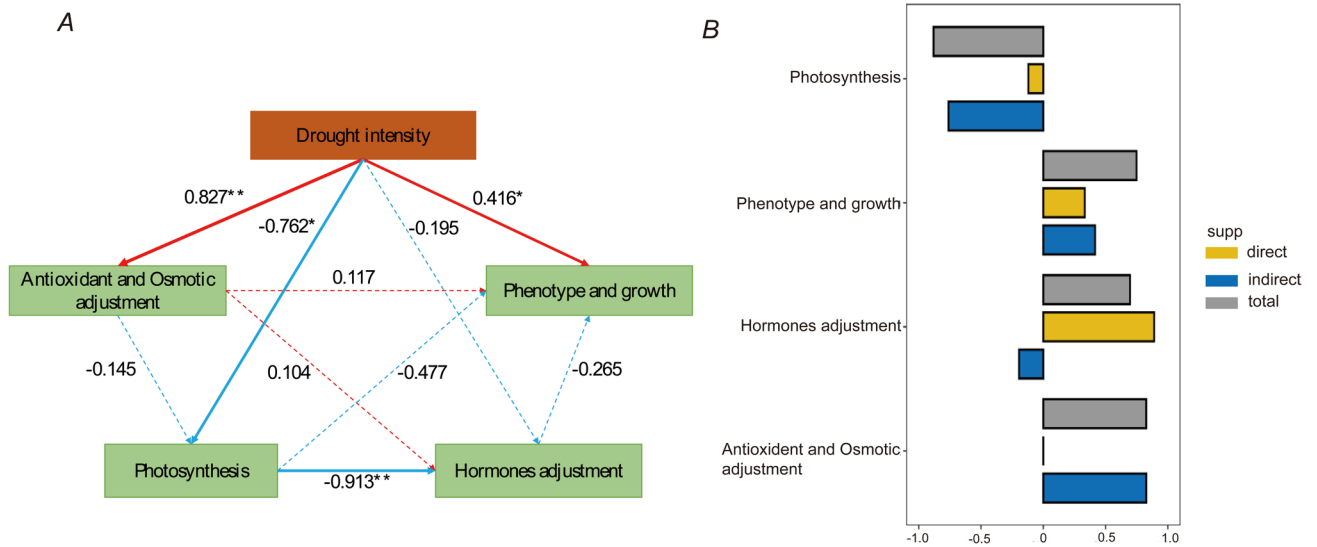


Fig. 4. (A) Structural equation model of drought responses in *Hippophae rhamnoides*, with standardized path coefficients. The model includes four latent constructs: Antioxidant and osmotic adjustment, Photosynthesis, Hormone adjustment, and Phenotype and growth. Arrows indicate standardized path coefficients. Solid lines indicate significant paths, whereas dashed lines indicate nonsignificant paths. Statistical significance is indicated by *P* values. (B) Direct, indirect, and total effects of drought intensity on the four functional modules. Blue, yellow, and gray bars represent direct, indirect, and total effects, respectively.

use efficiency (0.1239), all of which loaded positively. In contrast, chlorophyll content (−0.2773), stomatal limitation (−0.2569), transpiration rate (−0.3053), and maximum photochemical efficiency (−0.1793) loaded negatively. Higher construct values therefore reflected stronger carbon assimilation, greater stomatal openness, and higher photochemical energy use, together with lower stomatal limitation and lower chlorophyll content. After

combining multiple photosynthetic and photochemical metrics, net photosynthetic rate dominated the construct, whereas the remaining indicators primarily captured variation associated with stomatal limitation and pigment status.

(3) Hormone regulation: This construct was defined by four indicators (IAA, ZT, ABA, and GA). The largest positive loadings were observed for ZT (0.4492) and IAA

(0.4430), whereas ABA showed a moderate positive loading (0.1552). GA exhibited a relatively large negative loading (−0.5997). Thus, higher construct values corresponded to a hormonal balance characterized by relatively higher auxin and cytokinin signals and suppressed GA, with ABA providing a smaller positive contribution. In this sense, the construct reflects rebalancing among growth-related hormones and GA rather than a unidirectional change in any single hormone.

(4) Phenotype and growth: This construct was defined by plant height, leaf area, and total biomass. Plant height had the highest loading (−0.5572), followed by leaf area (−0.4721), whereas biomass also loaded negatively but more weakly (−0.1852). Higher construct values therefore corresponded to lower plant height, smaller leaf area, and reduced dry-matter accumulation. This implies that, across the synthesized literature, declines in plant height, canopy area, and biomass tended to occur in a coordinated manner. Accordingly, this construct appears to reflect an overall reduction in plant growth and canopy expansion under drought conditions, rather than contrasting dimensions of plant form or allocation. In this sense, variation within the latent construct represents a coordinated suppression of growth-related traits under drought stress.

Discussion

Stage-dependent antioxidant–osmotic adjustment and sources of heterogeneity: Drought-induced accumulation of reactive oxygen species (ROS) is a well-established topic in plant stress biology (Guo *et al.* 2007, Urbinati *et al.* 2020, Chen *et al.* 2024a). Accordingly, antioxidant enzyme activities and membrane lipid peroxidation have been widely used as core indicators of drought resistance (Gao *et al.* 2021, Liu *et al.* 2025b). The present synthesis shows that antioxidant enzymes should be treated as stage-dependent indicators rather than as general markers of drought resistance. SOD and POD increased from mild to moderate drought but declined sharply under severe drought, while CAT changed only slightly and became negative under severe stress. This nonmonotonic pattern helps explain why previous studies have reported inconsistent antioxidant responses in *H. rhamnoides*. In early or moderate drought, increased enzyme activity may reflect an active defense against ROS accumulation. Under severe or prolonged drought, however, enzyme activity may decline because cellular injury, substrate limitation, or protein instability exceeds the capacity of the antioxidant system. Therefore, a high SOD, POD, or CAT value at a single sampling point cannot be interpreted directly as stronger drought resistance. In comparison, membrane permeability, relative electrical conductivity, and leaf water content showed more stable response directions and are more suitable as priority indicators in cross-study evaluation. Their meta-analytic effect sizes were more stable and contributed more strongly to the SEM. These results indicate that cellular structural integrity and water retention impose more fundamental and robust constraints on drought responses in *Hippophae rhamnoides* (Gao *et al.* 2021, Zhang *et al.*

2021b). Accumulation of osmotic-adjustment compounds (e.g., Pro and soluble sugars) was also more consistent than changes in single enzyme activities and generally increased with stress severity (Zhang *et al.* 2020, Gao *et al.* 2021). Thus, antioxidant enzymes and osmolytes jointly buffer oxidative stress, but the capacity to maintain cellular structure and osmotic balance under dehydration is more likely to determine functional persistence in *Hippophae rhamnoides* (Blum 2017, Gao *et al.* 2021). From an applied perspective, equating high antioxidant enzyme activity with “strong drought resistance” may be risky; water-status and membrane-stability traits appear to be more appropriate priority indicators for screening and evaluation. This inference should be tested under standardized experimental conditions.

Stage-dependent shifts in photosynthetic limitation: stomatal vs. nonstomatal processes: Photosynthesis-related traits were highly sensitive to drought and provided one of the most consistent signals of drought response in *Hippophae rhamnoides*. The meta-analysis showed marked declines in net photosynthetic rate and transpiration under drought, accompanied by an overall reduction in stomatal conductance. By contrast, within the mild-to-moderate drought range, changes in chlorophyll content and photochemical parameters (e.g., maximum photochemical efficiency) tended to be smaller. This combination is consistent with an early response dominated by stomatal limitation (Chen *et al.* 2024a). In this phase, stomatal closure reduces transpirational water loss, and declines in photosynthesis are largely downstream consequences, while the photosynthetic apparatus has not yet exhibited clear structural impairment. As drought intensifies, some studies report reductions in chlorophyll content and PSII quantum efficiency, accompanied by increased nonphotochemical quenching and decreased electron transport efficiency (Liu *et al.* 2023). These patterns suggest that nonstomatal limitations strengthen and may become dominant, and that photosynthesis becomes increasingly constrained by structural impairment. At this stage, lower photosynthesis may reflect both damage to photosynthetic components and broader suppression of energy-demanding metabolism under severe water limitation (Chen *et al.* 2024a, Yang *et al.* 2024). Accordingly, declines in photosynthesis should not be interpreted simply as “failure” of drought resistance. The present synthesis supports a stage-dependent interpretation: under mild-to-moderate drought, reduced photosynthesis more closely reflects a regulatory strategy aimed at conserving water and resources; under severe drought, structural impairment and functional suppression co-occur, and the system shifts from predominantly active regulation to predominantly damage-driven limitation (Guo *et al.* 2007, Yang *et al.* 2024). In the SEM, the photosynthesis latent variable occupied an important integrative position linking drought intensity, water status, and hormonal balance, suggesting that photosynthesis may function as a key coordinating component within the present framework. Photosynthesis can act as an integrative hub because it is simultaneously constrained by water relations, membrane

condition, and metabolic demand. At the early stage of drought, declining leaf water status promotes stomatal closure, which reduces transpiration and CO₂ diffusion and leads to a decline in net photosynthetic rate. With increasing drought intensity, pigment loss, impaired electron transport, and enhanced energy dissipation indicate that nonstomatal limitation becomes more important. Once carbon assimilation is reduced, the supply of carbon skeletons and energy for growth, osmolyte synthesis, and repair processes is also constrained. Photosynthetic decline can therefore influence hormone balance and source–sink relationships, linking primary water deficit to growth suppression. In this sense, photosynthesis is not only a response variable but also an integrative state variable that reflects the combined effects of hydraulic stress, photochemical performance and carbon allocation. The SEM result is consistent with this interpretation, although it should be regarded as evidence of coordinated module-level association rather than proof of a one-way causal pathway. Nevertheless, most included studies relied on endpoint measurements under heterogeneous conditions and lacked standardized time series; consequently, uncertainty remains in distinguishing strategic downregulation from structural damage in the aggregated evidence base.

Reconfiguration of the hormonal network and coupling to photosynthesis and growth: An increase in ABA is widely regarded as a hallmark of plant drought responses (Jin *et al.* 2023, Yang *et al.* 2024), and *Hippophae rhamnoides* showed a broadly similar tendency. In the compiled evidence, ABA increased overall under drought, whereas hormones associated with cell elongation and division – including IAA, ZT – declined overall (Guo *et al.* 2007, Zhang *et al.* 2020). These shifts indicate that the hormonal network was reconfigured toward growth inhibition and stress tolerance, with the balance between ABA and growth-related hormones shifting in a drought-associated direction (Guo *et al.* 2007, He *et al.* 2016, Gao *et al.* 2021, Lin *et al.* 2025). However, the results do not support a simple linear pathway of “drought → ABA → drought resistance”. Both the correlation analysis and the structural equation model (SEM) suggested that the direct effect of drought intensity on the hormone latent variable was not prominent. Instead, the primary influence was expressed indirectly *via* photosynthesis and water status. Moreover, hormone measurements varied widely across studies in sampling time and tissue type, likely reducing cross-study consistency relative to water-status and photosynthetic traits (Guo *et al.* 2003, Yang *et al.* 2010, 2024; Gao *et al.* 2021, Lin *et al.* 2025). A more cautious interpretation is therefore warranted: hormonal regulation appears to be a network-level response integrating photosynthetic state, water balance, and growth demand, rather than a process dominated by any single hormone. Under this view, increases in ABA are better interpreted as feedback signaling in response to altered water status and photosynthetic function, rather than as a purely upstream control factor (Jin *et al.* 2023). Because the SEM was built from pooled effect sizes across studies and is constrained by sample size and heterogeneity, inferences about pathway

placement should be treated as hypothesis-generating and validated in controlled, unified experimental designs.

Theoretical contribution, applied implications, and

limitations: The main contribution of this study is not only the quantitative synthesis of drought-responsive traits, but also the identification of a functional hierarchy that organizes these traits into primary constraints, integrative physiological responses, buffering adjustments, and downstream growth outcomes. At the species level, this study integrated dispersed evidence on antioxidant and osmotic adjustment, photosynthesis, growth, and hormonal traits in *Hippophae rhamnoides* within a single functional framework. The combined meta-analysis and SEM indicate that drought responses are better described as a functional state shift – from “high investment in growth” toward a defense-prioritized state centered on maintaining water balance and structural stability – rather than sustained enhancement of any single trait class. Recent studies have further emphasized that plant drought responses are not expressed through isolated changes in single traits, but through coordinated adjustments among structural, physiological, and metabolic dimensions. For example, Kolyva *et al.* (2025) showed that water deficiency can induce integrated shifts across multiple functional trait domains, even though the magnitude and direction of specific responses may vary with developmental context. It supports the broader interpretive framework adopted here that drought adaptation is best understood as a coordinated response strategy jointly shaped by multiple functional modules. This perspective helps situate *Hippophae rhamnoides* drought adaptation within a broader functional–ecological framework (Kolyva *et al.* 2025). From an applied perspective, the results suggest that drought-resistance evaluation and screening should prioritize water-status and structural-stability traits (*e.g.*, leaf water content, membrane permeability, and relative electrical conductivity) and integrate these with photosynthetic and growth traits for a more comprehensive assessment. These indicators were more stable across studies and thus better suited for evaluation systems with broader generality. By contrast, single antioxidant enzyme activities and some hormonal traits remain valuable for mechanistic studies but are strongly influenced by experimental conditions and sampling schedules, limiting their reliability as standalone screening criteria. Several methodological limitations should be acknowledged. First, although drought intensity was standardized where possible, drought duration could not be standardized across studies. This limitation is important because stomatal conductance and transpiration may respond rapidly, whereas antioxidant enzymes, osmolytes, MDA, hormone levels, and growth traits may show delayed or biphasic responses. Therefore, some nonmonotonic responses observed in antioxidant, hormonal, and photosynthetic traits may partly reflect differences in exposure duration among the original studies. Second, variance information (*e.g.*, standard deviations) was missing for some traits and had to be approximated, introducing additional uncertainty. Third, the SEM was constructed from pooled effect

sizes across studies, and both sample size and variable homogeneity were constrained. Consequently, the inferred pathways are better viewed as hypothesis-generating and mechanistically informative rather than strict causal proof. Within these limits, the proposed functional hierarchy – water status and structural stability as foundational constraints, photosynthesis as a key integrative node, and antioxidants and hormones as supporting adjustments – offers an operational perspective on integrated drought adaptation in *Hippophae rhamnoides* and provides a methodological reference for cross-experiment syntheses in other drought-tolerant shrubs and trees.

Future research should combine standardized drought-gradient experiments with time-series sampling to separate early regulatory responses from delayed injury responses and cumulative growth effects. This proposed hierarchy is summarized in Fig. 5, which distinguishes primary stress constraints, stage-dependent buffering responses, integrative photosynthetic inhibition, hormonal reprogramming, and downstream growth consequences.

Practical use of indicator priorities in restoration and monitoring: These findings can be translated into a tiered indicator system for restoration practice. For early drought diagnosis, leaf water content, leaf water potential, membrane permeability, and relative electrical conductivity can be used as first-line indicators because

they respond directly to water deficit and cellular injury. For evaluating functional decline, net photosynthetic rate, stomatal conductance, transpiration rate, and chlorophyll-related traits should be monitored together, as they reflect whether drought has moved from reversible stomatal regulation toward stronger nonstomatal limitation. For longer-term restoration assessment, plant height, leaf area, and biomass remain necessary because they indicate whether physiological adjustment has been sufficient to maintain growth. Antioxidant enzymes and hormones are more suitable for mechanistic experiments or controlled screening programs, but they should not be used alone as field indicators because their responses are strongly affected by drought stage, tissue type, and sampling time.

Conclusions and recommended indicator priorities: This study established a species-level functional hierarchy of drought adaptation in *Hippophae rhamnoides* by integrating evidence from 49 drought experiments across 12 provinces in China, comprising 3,855 treatment–control comparisons, with meta-analysis and structural equation modeling. The results demonstrated that deterioration in water status and membrane stability represents the most fundamental and persistent constraint under drought stress, whereas photosynthesis functions as the central integrative hub linking water deficit, hormonal reprogramming, and growth suppression. In contrast, antioxidant defense and

Conceptual flow chart of the proposed functional hierarchy of drought adaptation in *H. rhamnoides*

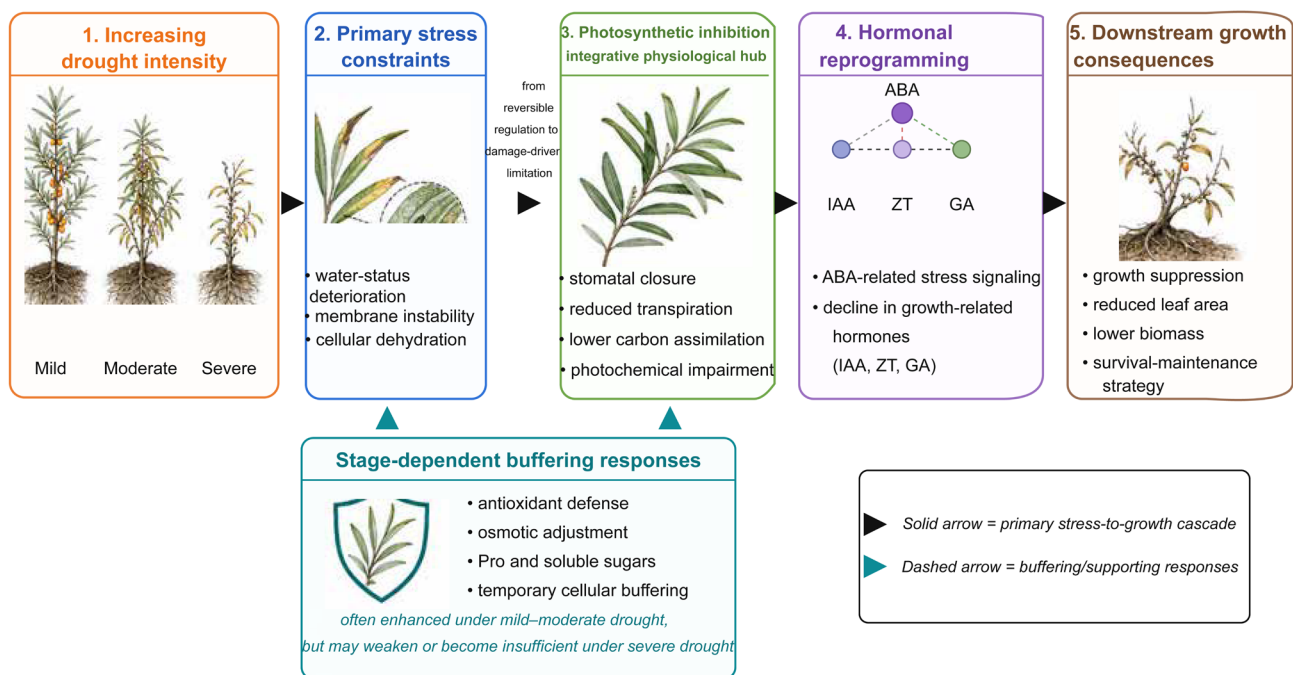


Fig. 5. Conceptual flow chart of the proposed functional sequence of drought responses in *Hippophae rhamnoides*. Increasing drought intensity is associated with deterioration in water status and membrane stability, followed by photosynthetic inhibition, hormonal adjustment, and reduced growth. Antioxidant defense and osmotic adjustment are presented as stage-dependent buffering responses that may contribute to stress adjustment under mild to moderate drought but decline under severe drought. ABA – abscisic acid; IAA – indole-3-acetic acid; ZT – zeatin; GA – gibberellin; Pro – proline.

osmotic adjustment acted primarily as stage-dependent buffering responses rather than continuously enhanced resistance mechanisms. Collectively, these responses indicate a coordinated shift from growth maintenance to defense prioritization as drought intensity increases. Beyond clarifying the physiological basis of drought resistance in *Hippophae rhamnoides*, this study provides a quantitative framework for identifying key drought-response indicators and offers practical implications for shrub selection, vegetation restoration, and water-management strategies in arid and semi-arid regions. Nevertheless, these conclusions should be interpreted with caution because drought duration was not standardized across the original studies. As a result, the estimated effects mainly represent endpoint responses under different reported drought intensities and may not fully separate the effects of stress intensity from those of exposure time. Future research should combine standardized drought-gradient experiments with time-series sampling designs to distinguish early regulatory responses from delayed injury responses and cumulative growth effects.

References

- Ai Y.: [Study on carbon allocation strategies of the drought-adapted sandy plants in southeastern Tibet.] PhD Thesis. Tibet University, Lhasa 2024. [In Chinese]
- Blum A.: Osmotic adjustment is a prime drought stress adaptive engine in support of plant production. – *Plant Cell Environ.* **40**: 4-10, 2017.
- Cai H., Wu F., Yang W.: [Effects of drought stress on the photosynthesis of *Salix paraqpleisia* and *Hippophae rhamnoides* seedlings.] – *Acta Ecol. Sin.* **31**: 2430-2436, 2011. [In Chinese]
- Chen H., Chen X., Li X. *et al.*: Growth and physiological response of Yulu *Hippophae rhamnoides* to drought stress and its omics analysis. – *Plant Signal. Behav.* **19**: 2439256, 2024a.
- Chen H., Chen X., Liu C. *et al.*: [Growth and physiological responses of *Hippophae rhamnoides* "Yulu" and *H. rhamnoides* "Shenqiuhong" to drought stress.] – *J. Northwest For. Univ.* **39**: 104-112, 2024b. [In Chinese]
- Chen Z., Yuan M., Cao Q. *et al.*: [Effects of soil water content on stomatal conductance and leaf water potential of *Medicago sativa* and *Hippophae rhamnoides*.] – *Soil Water Conserv. Sci.* **17**: 37-43, 2019. [In Chinese]
- Cheng P., Tang D., Wang K. *et al.*: [Responses and drought resistance evaluation of 12 species of plants to drought stress and rehydration in the Northern Piedmont, Tianshan Mountains.] – *Acta Bot. Bor.-Occident. Sin.* **44**: 691-705, 2024. [In Chinese]
- Cui D.: [Study on drought resistance physiological and ecological characteristics of *Hippophae rhamnoides* seedlings.] Master Thesis. Northeast Forestry University, Harbin 2010. [In Chinese]
- Cui D., Wang S., Ma Y.: [Study on physiological and ecological characteristics of Russian big-fruit seabuckthorn seedlings under drought stress.] – *Northern Hortic.* **16**: 15-19, 2010. [In Chinese]
- Cui Q.: [Effects of water deficit stress on membrane permeability and malondialdehyde (MDA) content in *Hippophae rhamnoides*.] – *Mod. Agric. Sci. Technol.* **11**: 139, 145, 2017. [In Chinese]
- Du C., Ni X., Yan M. *et al.*: Physiological and transcriptome analysis reveals the mechanism of *Gymnocarpus przewalskii* response to drought stress. – *BMC Plant Biol.* **25**: 155, 2025.
- Fang J., Wu F., Yang W. *et al.*: Effects of drought on the growth and resource use efficiency of two endemic species in an arid ecotone. – *Acta Ecol. Sin.* **32**: 195-201, 2012.
- Fang L., Zeng Z., Jia Q. *et al.*: Physiological response and phytoremediation potential of dioecious *Hippophae rhamnoides* inoculated with arbuscular mycorrhizal fungi to Pb and Zn pollution. – *Front. Plant Sci.* **14**: 1321885, 2024.
- Gan T., He Z., Xu D. *et al.*: Modeling the potential distribution of *Hippophae rhamnoides* in China under current and future climate scenarios using the biomod2 model. – *Front. Plant Sci.* **16**: 1533251, 2025.
- Gao G.: [The physiological and molecular mechanism of seabuckthorn (*Hippophae rhamnoides*) in response to drought stress and rehydration.] PhD Thesis. Chinese Academy of Forestry, Beijing 2018. [In Chinese]
- Gao G., Lv Z., Zhang G. *et al.*: An ABA-flavonoid relationship contributes to the differences in drought resistance between different *Hippophae rhamnoides* subspecies. – *Tree Physiol.* **41**: 744-755, 2021.
- Gao J.: [Effects of different soil water contents on sulfur metabolism of *Robinia pseudoacacia* and *Hippophae rhamnoides* L. subsp. *sinensis*.] PhD Thesis. Northwest A&F University, Yangling 2010. [In Chinese]
- Gao L., Yang J., Liu R.-X.: [Effects of soil moisture levels on photosynthesis, transpiration, and moisture use efficiency of female and male plants of *Hippophae rhamnoides* ssp. *sinensis*.] – *Acta Ecol. Sin.* **29**: 6025-6034, 2009. [In Chinese]
- Guo W., Li B., Zhang X., Wang R.: Architectural plasticity and growth responses of *Hippophae rhamnoides* and *Caragana intermedia* seedlings to simulated water stress. – *J. Arid Environ.* **69**: 385-399, 2007.
- Guo W.-H., Bo L., Huang Y.-M. *et al.*: Effects of different water stresses on eco-physiological characteristics of *Hippophae rhamnoides* seedlings. – *Acta Bot. Sin.* **45**: 1238-1244, 2003.
- Han R., Li L., Liang Z. *et al.*: [*Hippophae rhamnoides* membrane-lipid peroxidation system under drought stress.] – *J. Northwest For. Univ.* **4**: 1-5, 2002. [In Chinese]
- He C., Li M., Luo H. *et al.*: [Comprehensive evaluation on drought resistance of *Hippophae rhamnoides*.] – *For. Sci. Res.* **28**: 634-639, 2015. [In Chinese]
- He C.Y., Zhang G.Y., Zhang J.G. *et al.*: Physiological, biochemical, and proteome profiling reveals key pathways underlying the drought stress responses of *Hippophae rhamnoides*. – *Proteomics* **16**: 2688-2697, 2016.
- Hedges L.V., Gurevitch J., Curtis P.S.: The meta-analysis of response ratios in experimental ecology. – *Ecology* **80**: 1150-1156, 1999.
- Hou L.: [The study on changing regularity of physiological and biochemical indexes under drought stress of *Hippophae rhamnoides*.] PhD Thesis. Inner Mongolia Agricultural University, Hohhot 2007. [In Chinese]
- Hu J., Liu L., Lu S. *et al.*: [Effect and mechanism of supplementary application of exogenous calcium on growth of *Hippophae rhamnoides* under drought stress.] – *Shanxi For. Sci. Technol.* **50**: 24-26, 32, 2021. [In Chinese]
- Jin Y., Hao G., Hammond W.M. *et al.*: Aridity-dependent sequence of water potentials for stomatal closure and hydraulic dysfunctions in woody plants. – *Glob. Change Biol.* **29**: 2030-2040, 2023.
- Kanukula R., Page M.J., Turner S.L., McKenzie J.E.: Identification of application and interpretation errors that can occur in pairwise meta-analyses in systematic reviews of interventions: a systematic review. – *J. Clin. Epidemiol.* **170**: 111331, 2024.
- Kolyva F., Adamakis I.-D.S., Gkikas D. *et al.*: Effects of elevated

- temperature and water deficiency on functional traits of *Vitis vinifera* L. cv. Assyrtiko leaves. – *Plants-Basel* **14**: 2463, 2025.
- Li L.: [Studies on the germination and drought-resistance physiological mechanism in seabuckthorn under water deficit situation.] PhD Thesis. Northwest A&F University, Yangling 2001. [In Chinese]
- Li L., Liang Z., Han R.: [Effect of soil drought on the growth and water-use efficiency of *Hippophae rhamnoides*.] – *Acta Bot. Bor.-Occident. Sin.* **2**: 88-94, 2002. [In Chinese]
- Li L., Liang Z., Wei Y. *et al.*: [Changes in endogenous hormones of *Hippophae rhamnoides* in dormancy stage and sprout period under soil water stress and exogenous GA3 regulation.] – *J. Northwest For. Univ.* **2**: 10-14, 2001. [In Chinese]
- Li L., Zhao J., Han R.: [Effect of drought stress on antioxidant enzymes in *Hippophae rhamnoides*.] – *J. Yantai Univ.* **1**: 30-35, 2006. [In Chinese]
- Li Y., Chen J., Luo Y.: [Effect of drought on growth and physiological traits of two species of *Hippophae rhamnoides*.] – *J. Fujian For. Sci. Technol.* **44**: 10-15, 2017. [In Chinese]
- Lin F., Cai Y., Yang S., Yang Y.: Alternative splicing events and ABA hormone regulation in drought response of *Hippophae gyantsensis* L. – *Genes* **16**: 350, 2025.
- Liu G., Yue L., Liu C. *et al.*: [Effects of drought stress on photosynthetic fluorescence parameters of *Hippophae rhamnoides* ssp. *Russica* seedlings.] – *Northern Hortic.* **22**: 16-23, 2022. [In Chinese]
- Liu J., Zhang R., Zhang G. *et al.*: Effects of soil drought on photosynthetic traits and antioxidant enzyme activities in *Hippophae rhamnoides* seedlings. – *J. Forestry Res.* **28**: 255-263, 2017.
- Liu M., Zhao G., Huang X. *et al.*: Candidate regulators of drought stress in tomato revealed by comparative transcriptomic and proteomic analyses. – *Front. Plant Sci.* **14**: 1282718, 2023.
- Liu Q., Ye G., Ma Y.: Development and application of microsatellite markers in *Hippophae rhamnoides* subsp. *sinensis* Rousi (*Hippophae rhamnoides* L.) based on transcriptome sequencing. – *Front. Genet.* **15**: 1373028, 2024.
- Liu R.: [Analysis on the ecological adaptability to drought stress of native and introduced *Hippophae* species.] PhD Thesis. Inner Mongolia University, Hohhot 2005. [In Chinese]
- Liu R., Yang J., Gao L.: [Changes in contents of proline, soluble saccharin, and endogenous hormone in leaves of Chinese *Hippophae rhamnoides* and Russian *Hippophae rhamnoides* under different soil water content.] – *J. Soil Water Conserv.* **3**: 148-151+169, 2005. [In Chinese]
- Liu R., Yang J., Gao L.: [Effects of varying soil water conditions on photosynthetic and transpiration rates of Chinese and Russian *Hippophae rhamnoides*.] – *Acta Sci. Nat. Univ. NeiMongol* **2**: 163-169, 2006. [In Chinese]
- Liu Z., Chen Y., Murtaza G. *et al.*: Exogenous application of limonene alleviates drought-induced oxidative stress in tobacco plants. – *Ind. Crop. Prod.* **234**: 121639, 2025b.
- Liu Z., Zhao Y., Ma Y.: [Effects of drought stress on physiological traits of Chinese *Hippophae rhamnoides* seedlings.] – *Agric. Disaster Res.* **15**: 131-133, 2025a. [In Chinese]
- Luo Y.Q., Hui D.F., Zhang D.Q.: Elevated CO₂ stimulates net accumulations of carbon and nitrogen in land ecosystems: a meta-analysis. – *Ecology* **87**: 53-63, 2006.
- Lyu Z., Zhang G., Song Y. *et al.*: Transcriptome and DNA methylome provide insights into the molecular regulation of drought stress in sea buckthorn. – *Genomics* **114**: 110345, 2022.
- Ma Y., Gao Y., Xiang Q. *et al.*: [The response to drought stress and drought-resistance evaluation of two wild *Hippophae rhamnoides* species in Qinghai Province.] – *Hubei Agric. Sci.* **53**: 3827-3831, 2014. [In Chinese]
- Ma Y., Gao Y., Zhou Z. *et al.*: [Effects of drought stress on physiological characteristics of *Hippophae neurocarpa*.] – *J. Qinghai Univ.* **31**: 1-5, 2013b. [In Chinese]
- Ma Y., Liu P., Gao Y. *et al.*: [Changes of osmoregulation substances and protective enzyme activities of *Hippophae neurocarpa* under drought stress.] – *J. Gansu Agric. Univ.* **48**: 100-104, 2013a. [In Chinese]
- Ma Y., Liu P., Zhou Z. *et al.*: [Effect of drought stress on the physiological index and protective enzyme activity of wild *Hippophae rhamnoides* subsp. *sinensis* in Qinghai Province.] – *Hubei Agric. Sci.* **52**: 5771-5775, 2013c. [In Chinese]
- Mou J.: [Effects of drought and salt stresses on photosynthetic characteristics of the four greening species.] PhD Thesis. Lanzhou University, Lanzhou 2020. [In Chinese]
- Nan J., Danzeng N., Lin L.: [Photosynthetic characteristics of three species of *Hippophae* in Tibet under drought stress.] – *Plateau Agric.* **3**: 606-614, 2019. [In Chinese]
- Nan J., Yang G., Wu T., Lin L.: [Comparison of drought resistance of three *Hippophae* species in Tibet.] – *J. Northwest A&F Univ.* **49**: 37-47, 2021. [In Chinese]
- Pei B., Zhang G.C., Zhang S.Y. *et al.*: [Effects of soil drought stress on photosynthetic characteristics and antioxidant enzyme activities in *Hippophae rhamnoides* Linn.] – *Acta Ecol. Sin.* **33**: 1386-1396, 2013. [In Chinese]
- Song X.: [Effects of drought stress on SOD and POD activities in four *Hippophae rhamnoides* varieties.] – *Green Sci. Technol.* **3**: 38-39, 2016. [In Chinese]
- Sun X., Jia N., He P. *et al.*: [Effects of drought stress on the growth and development of *Hippophae rhamnoides* seedlings in northern Shaanxi.] – *Northeast Agric. Sci.* **43**: 16-20, 2018. [In Chinese]
- Tang D., Cheng P., Yang J. *et al.*: [Physiological responses of piedmont vegetation on the northern slope of the Tianshan Mountains to drought stress.] – *Arid Zone Res.* **38**: 1683-1694, 2021. [In Chinese]
- Tang T., Ge J., Shi H. *et al.*: Drought frequency, intensity, and exposure have increased due to historical land use and land cover changes. – *Commun. Earth Environ.* **6**: 398, 2025.
- Tian J.: [Effect of drought stress on *Hippophae rhamnoides* seedling growth.] – *Rural Sci. Technol.* **15**: 97-99, 2024. [In Chinese]
- Urbinati G., Nota P., Frattarelli A. *et al.*: Morpho-physiological responses of sea buckthorn (*Hippophae rhamnoides*) to NaCl stress. – *Plant Biosyst.* **154**: 827-834, 2020.
- Wang G.: [Study on the relationship between morphological as well as physiological indexes and drought resistance of seabuckthorn.] PhD Thesis. Inner Mongolia Agricultural University, Hohhot 2006. [In Chinese]
- Wang H.: [Physiological responses of *Hippophae rhamnoides* varieties in the Fuxin region to drought treatment.] – *Mod. Agric. Sci. Technol.* **14**: 151, 154, 2018. [In Chinese]
- Wu Q., Zhang G., Pei B., Fang L.: [Physiological and biochemical responses to different soil drought stress in three tree species.] – *Acta Ecol. Sin.* **33**: 3648-3656, 2013. [In Chinese]
- Xia Y.: [The effects of exogenous calcium on the growth and physiological characteristics of *Hippophae rhamnoides* under drought.] PhD Thesis. Shenyang Agricultural University, Shenyang 2024. [In Chinese]
- Xu C., Zhao Z., Chen M.: [Effects of water stress on physiological indicators in tree roots of five varieties.] – *J. Northwest A&F Univ.* **37**: 109-114, 2009. [In Chinese]
- Xu G.: [Responses of different *Hippophae rhamnoides* subsp. *sinensis* populations to drought stress and analysis of drought-induced proteins.] PhD Thesis. Graduate School of Chinese

- Academy of Sciences, Beijing 2007. [In Chinese]
- Xu J., Chen H., Liu F. *et al.*: [Comprehensive evaluation on drought resistance of different *Hippophae rhamnoides* in Ulan Buh Desert.] – Bull. Soil Water Conserv. **35**: 88-94, 2015. [In Chinese]
- Yang J., Yang B., Yang J. *et al.*: Leaf flavonoids in Chinese sea-buckthorn (*Hippophae rhamnoides* subsp. *sinensis* Rousi) and their response to environmental gradients across northern China. – ScienceAsia **50**: 1-10, 2024.
- Yang Y., Yao Y., Zhang X.: Comparison of growth and physiological responses to severe drought between two altitudinal *Hippophae rhamnoides* populations. – Silva Fenn. **44**: 603-614, 2010.
- Zhang D., Ma H., Zhang X. *et al.*: [Changes in proline content and expression patterns of key regulatory genes in *Hippophae rhamnoides* subsp. *sinensis* under drought stress.] – Mol. Plant Breed. **19**: 7596-7603, 2021a. [In Chinese]
- Zhang G., Lv Z., Diao S. *et al.*: Unique features of the m⁶A methylome and its response to drought stress in sea buckthorn (*Hippophae rhamnoides* Linn.). – RNA Biol. **18**: 794-803, 2021b.
- Zhang H., Zha T., Li X. *et al.*: [Photosynthetic and physiological responses of six shrub species to drought stress in northern Hebei Province.] – Chin. J. Ecol. **40**: 352-362, 2021c. [In Chinese]
- Zhang L.: [Study on drought adaptability between Russian and Chinese *Hippophae rhamnoides*.] – For. By-Prod. Spec. China **3**: 25-27, 2010. [In Chinese]
- Zhang R.: [Ecological adaptability of typical woody plant seedlings under drought stress in mining areas of the Loess Plateau.] PhD Thesis. Shanxi University, Taiyuan 2015. [In Chinese]
- Zhang T., Gao G., Liu J. *et al.*: Transcripts and ABA-dependent signaling in response to drought stress in *Hippophae rhamnoides* L. – Trees **34**: 1033-1045, 2020.

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